

Henry Yu

New York, NY | henry.yu@stonybrook.edu | 917-831-2139 | [linkedin.com/in/henryyu0505](https://www.linkedin.com/in/henryyu0505) | github.com/henhen0505

EDUCATION

Stony Brook University

Stony Brook, New York

Bachelor of Science in Computer Science, Focus in Algorithms and Systems

May 2027

Double major in Applied Mathematics and Statistics, Concentration in Probability and Optimization

Relevant Coursework: Data Structures, Algorithms, Analysis of Algorithms, Graph Theory, Systems Fundamentals, Computer Networks, Theory of Computation, Linear Algebra

PROJECTS

SEC Filing Evaluation Harness | *Python, RAG, Anthropic Claude API, pandas, pytest*

August 2026

- Built an evaluation harness for numerical accuracy on SEC filings against the open **LEDGER** benchmark, backed by an **80+** test suite in **pytest**.
- Generated **3,700+** questions on computed metrics (margins, ratios, YoY growth) from **SEC EDGAR XBRL** ground truth, extending the benchmark past extraction alone.
- Ran a diagnostic evaluation over **25** questions across **~20** companies: **72%** of answers found the right figure at the wrong scale, masked by a **4%** raw accuracy score.

Playlist – Full-Stack Web App | *React, Node.js, Express, MongoDB, PostgreSQL, Sequelize*

October 2025

- Built a **MERN** app with **JWT** sessions and ownership checks on auth, playlist, and song **REST** endpoints.
- Built a database abstraction layer with **MongoDB** and **PostgreSQL** implementations behind one interface, switchable by environment flag with no controller changes.
- Added multi-field catalog search over the **YouTube Data API** and a **jsTPS** undo/redo stack for reversible edits, tested with **Vitest**.

Market Data Engine | *C++, MySQL, GoogleTest, NASDAQ ITCH 5.0*

July 2026 – August 2026

- Wrote a **C++** parser for the **NASDAQ ITCH 5.0** binary protocol, reconstructing order books across **5** symbols from **55,748** raw messages at **~550 MB/sec** throughput, verified by **61 GoogleTest** cases.
- Built a **MySQL** persistence layer with batch ingestion and snapshot capture, plus analytics queries (VWAP, rolling volatility, spread/depth, order-to-trade ratio) over real NASDAQ tick data.

WORK EXPERIENCE

BetterTherapy

New York, NY

Software Engineering Intern

May 2025 – August 2025

AI health-tech startup on the Bittensor decentralized inference network

- Designed a **SQLAlchemy** data layer over **SQLite** for an automated LLM evaluation pipeline: schema with a reusable timestamp mixin, cascading request/response relationships, and a session decorator.
- Wrote atomic upserts (ON CONFLICT DO UPDATE) for a node blacklist, incrementing violation counts safely under concurrent writes.
- Avoided N+1 round trips with **selectinload** eager loading, added bulk inserts, and polled the **OpenAI Batch API** on 24-hour windows.

ReWild Long Island

Long Island, NY

Software Engineer, Dev Team

August 2026 – Present

- Architected and containerized a full-stack backend (**FastAPI**, **PostgreSQL 16**, pgvector) with **Docker Compose** and **Alembic** migrations; designed an 11-table schema with composite keys, custom enums, and a many-to-many crosswalk for 100+ Long Island communities.

Stony Brook University

Stony Brook, New York

Undergraduate Teaching Assistant

August 2026 – Present

CSE 260: Honors Data Structures & Programming Abstractions

- Run lab sections for **~80** students, guiding implementation and analysis of data structures and programming abstractions.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Java, C/C++, Bash, HTML/CSS

Frameworks/Tools: React, Node.js, Express, FastAPI, MongoDB, PostgreSQL, MySQL, SQLAlchemy, Git/GitHub, REST APIs, JWT Authentication, Linux, AWS, Docker, LLM APIs, GoogleTest

Libraries: pandas, NumPy, statsmodels, matplotlib

Concepts: Data Structures & Algorithms, OOP, Full-Stack Web Development, System Design, API Design, Event-Driven Architecture, State Management, Testing (pytest)